

Armaan Rai

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EDUCATION

Shiley-Marcos School of Engineering, University of San Diego

B.S. Computer Science, Minor: Mathematics

Expected May 2027

RELEVANT COURSEWORK

Computational Problem Solving

Programming Abstractions and Methodologies

Object Oriented Programming with C++

Structured Programming

SKILLS

Programming Languages: C++, Python

Operating Systems: Microsoft Windows, iOS, Visual Studio Code, Adobe

PROJECT

Design WORDY Program Project

Spring 2025

- Designed and implemented a Wordle-style game using Python and the Tkinter GUI library, applying the Model-View-Controller (MVC) architecture to separate game logic, interface, and control flow.
- Built interactive game features including dynamic keyboard input, guess validation, and animated letter color feedback based on correctness using event-driven programming and custom exception handling.
- Incorporated file I/O for user input logging, developed modular, testable code with pytest and assert-based testing to ensure functionality and maintainability.

Design BLOCKY Program Project

Spring 2025

- Designed and implemented a recursive tree-based board game in Python, modeling hierarchical data using a quadtree structure and rendering interactive game states with Pygame.
- Applied object-oriented programming and inheritance to build a flexible class system supporting multiple player types (human, random, smart) each with unique scoring logic and behavior.
- Developed and optimized algorithms for goal-based scoring using blob detection and 2D grid transformations, enabling real-time feedback.

EXPERIENCE

Data Entry, Customer Service Representative

Sacramento, CA

7-Eleven #24147

January 2020

- Developed and maintained structured financial records for four business locations using spreadsheets and database tools, ensuring accuracy in daily sales, expense tracking, and bank reconciliations.
- Assisted in financial data analysis for budgeting, inventory cost evaluation, and monthly reporting; contributed insights to improve profitability and operational efficiency.
- Collaborated with a functional team to streamline data management, ensure accurate financial systems, and support reporting and compliance through effective communication and coordination.

ACTIVITIES, LEADERSHIP, and ASSOCIATIONS

Member: USD Computer Science

2025- Present

Member: USD Society of Asian Scientists and Engineers

2025- Present

Participant: USD Competitive Intramural Soccer 7v7

2025- Present

Member: SASE - Society of Asian Scientists and Engineers

2025- Present